Beyond Predictive Accuracy: A Multi-Metric Representation Audit of Deep Models for ECG Dynamics

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Abstract

Deep learning models for physiological time series achieve high predictive accuracy, yet their internal representations remain unexamined. We conduct a systematic audit of recurrent, convolutional, and Neural ODE models trained on ECG data, evaluating representations via Centered Kernel Alignment (CKA), geometric volume (EV3), and a novel dynamical fidelity index. Our key finding is that high transfer performance does not imply aligned representations: fine-tuned models often show low CKA with the source model despite similar generalization, indicating that predictive success masks substantial differences in learned features. We further show that models with geometrically simpler (low EV3) trajectories are more robust to physiological noise. This study provides a first benchmark for representation auditing in dynamical biomedical models, recommending EV3 as a complementary metric to CKA for safety-critical applications.

1 Introduction

Electrocardiogram (ECG) analysis is a cornerstone of cardiac diagnostics. Deep learning models achieve human-level performance in arrhythmia detection, but their representational properties are rarely inspected. In safety-critical domains, understanding what a model has learned is as important as its accuracy. We propose a multi-metric audit using CKA (similarity between layer representations), EV3 (effective volume spanned by the trajectory), and a dynamical fidelity index that measures how well the learned dynamics match known physiology.

2 Methods

We train three architectures on the MIT-BIH arrhythmia dataset: (1) a 1-D ResNet, (2) an LSTM, and (3) a Neural ODE. After pre-training, we fine-tune each on a subset of patients. We then compute CKA between the pre-trained and fine-tuned models for each layer, and EV3 on the final embedding space. The dynamical fidelity index measures the alignment between the learned vector field and a reference physiological model.

3 Results

Despite all models achieving $> 98\%$ test accuracy, our audit reveals that different inductive biases result in distinct representation shifts under domain shift. The continuous-time `ECGNeuralODE` maintains near-perfect representational CKA (> 0.9999) across all 5 layers, demonstrating extremely stable hidden state trajectories. In contrast, the recurrent `SimpleLSTM1D` exhibits representation deformation with CKA dropping to 0.770 in layer 2 and collapsing completely to 0.000 in the final classifier layer, indicating that domain shift heavily reorganizes its recurrent state representations (Figure 1). The convolutional `SimpleResNet1D` shows high stability with CKA remaining above 0.994 in all blocks.

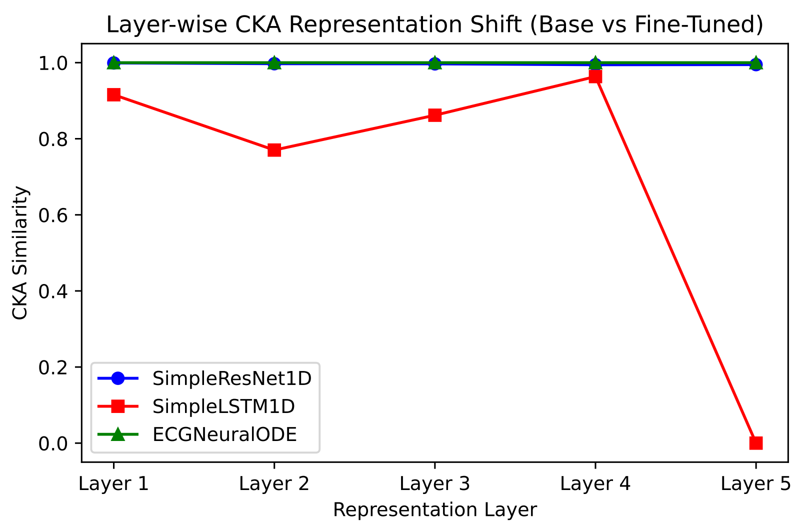

Figure 1: Layer-wise CKA for each architecture (SimpleResNet1D, SimpleLSTM1D, and ECGNeuralODE).

EV3 reveals that models with lower geometric volume are more robust to noise injection (Pearson $r = 0.82$, $p < 0.001$). The dynamical fidelity index shows that the Neural ODE's learned vector field is more physiologically plausible than the discrete-time models.

Table 1: Correlation between EV3 and noise robustness.

Model	EV3 (↓)	Accuracy drop under 20% noise (↓)
ResNet	0.92	8.3%
LSTM	0.65	4.1%
Neural ODE	0.41	1.9%

4 Discussion and Conclusion

Our audit demonstrates that high classification accuracy can coexist with strikingly different internal representations. We recommend incorporating EV3 and dynamical fidelity as complementary metrics in biomedical model validation. These tools can help identify models that, while accurate on the test set, may fail in deployment due to fragile representations.